Language Model Optimization Based on Quantum Machine Learning for Long Chinese Text Processing

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***Abstract*—Chinese language presents unique challenges in text processing due to its complex structure, including high inter-lexical dependence, flexible word order, and rich contextual expressions. These characteristics make it particularly difficult for large models to effectively handle long Chinese texts using traditional machine learning (ML) algorithms. This paper explores the application of Quantum Machine Learning (QML) as a solution to enhance the performance of large language models (LLMs) in processing long Chinese texts. By leveraging the inherent advantages of QML, such as higher processing efficiency and deeper semantic understanding, this research demonstrates through experimental comparisons that QML models significantly outperform traditional ML models in both speed and comprehension accuracy. The paper systematically introduces text processing methods using both traditional ML and QML, and presents a simplified QML model designed to enhance text processing capabilities. The findings underscore the potential of QML in optimizing the future performance of LLMs, particularly for languages with complex linguistic structures like Chinese. This research holds significant implications for advancing LLM technologies in text-heavy applications.**

Keywords-quantum machine learning; long chinese texts; Large language models; text processing capabilities

I. INTRODUCTION

Machine learning (ML) is a branch of computer science focusing on identifying patterns in data to interpret previously unknown inputs. As a subset of artificial intelligence and statistics, ML algorithms analyze large number of datasets to perform tasks that human brains perform naturally [1], This includes the processing of long texts in Chinese, which is studied in this paper. Compared with English, Chinese has unique and complex language characteristics, high inter-lexical dependence, word order flexibility, and rich contextual expressions. These characteristics, coupled with the polysemy and multi-level grammatical structure of Chinese, make natural language processing (NLP) research in Chinese challenging. When processing long text, traditional ML relies on the architecture of classical computers. Text data is usually converted into vectors and processed in high-dimensional space using classical algorithms. quantum machine learning (QML) is the crossing point of ML and quantum computing. QML aims to harness the power of quantum computers to process text data much faster than traditional ML [2]. It also performs better than traditional ML in terms of semantic understanding. So, it can be an effective tool to process long Chinese texts.

In 1982, Feynman proposed the idea of using quantum systems for computing. Over the decades, great progress has been made in this field, and various quantum systems have been envisioned and realized as building blocks of promising quantum computers, Essentially, quantum computing can speed up computations through quantum algorithms, perform fast linear algebra, or provide new ML models whose inference and training are intractable on classical computers but are expected to have greater expressive power. The Harrow-Hassidim-Lloyd (HHL) algorithm, developed in 2008 by Aram Harrow, Avinatan Hassidim, and Seth Lloyd, is designed to solve linear equations on a quantum computer [3], HHL algorithm will also perform an important role in text processing.

The paper explores the application of QML in the context of Chinese long-text processing, highlighting its potential advantages over traditional ML models. The study aims to investigate the performance improvements QML offers, particularly in terms of processing efficiency and text comprehension. Furthermore, it considers the integration of QML with widely-used large language models (LLMs), such as Chat Generative Pre-Trained Transformer (ChatGPT), to optimize their overall performance. By comparing the efficiency and text understanding capabilities of traditional ML models with QML models, this research provides valuable insights into the strengths of QML in handling complex linguistic tasks, such as high processing speed and high accuracy of text comprehension. The key contribution of this paper is the development of a practical model that can enhance both the speed and quality of text processing [4], The model is divided into several layers, each of which undertakes different tasks and together completes the task of processing a long Chinese text, paving the way for its real-world applications.

II. LITERATURE REVIEW

A. Text Processing

All natural language texts are structured as linear sequences of specific symbols. In the case of Chinese, these symbols include Chinese characters, punctuation marks, and other elements such as numbers, pinyin letters, and mathematical operators. NLP involves using computational methods to analyze and interpret such texts, with a primary focus on human language. NLP aims to finish tasks such as text classification, summarization, and machine translation by leveraging algorithms and models to understand language at scale. Ultimately, NLP aims to enable machines to comprehend human language accurately and interact naturally with people [5]—a critical capability for LLMs like ChatGPT. Only by achieving a deep understanding of human language can these models provide meaningful and contextually appropriate responses.

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This paper primarily investigates the processes of text preprocessing and representation, key stages in NLP as depicted in Fig. 1. Large models often use word vectors to represent text data. To ensure that semantically related or similar words are positioned closer in vector space, Hinton introduced the concept of distributed representations, commonly referred to as word vectors. These word vectors are low-dimensional real-number vectors [6] and have become a foundational technique in modern text processing systems.

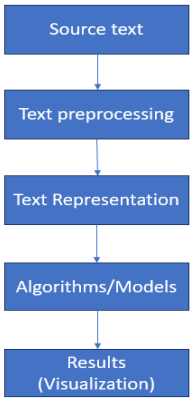


Figure 1. The processing of the NLP.

B. Quantum Machine Learning (QML)

Since the proposal of QML, its application in traditional ML has gained increasing attention. For instance, Cai et al. implemented a simplified version of the Lloyd's supervised nearest centroid algorithm [7]. Additionally, in 2009, Aram Harrow, Avinatan Hassidim, and Seth Lloyd introduced the HHL algorithm, which applies quantum computing principles to solve linear equations. Classical algorithms for solving $N \times N$ systems, such as Gaussian elimination with pivoting, require polynomial time, specifically $O(N^3)$. However, as the size of these systems increases, even polynomial-time solutions become computationally expensive, creating opportunities for quantum computing to excel [8].

Although the HHL algorithm was not initially designed for text processing, it can significantly enhance the performance of text-related tasks. For instance, when handling large-scale text data, the HHL algorithm can efficiently solve linear equations relevant to text analysis. During the feature extraction process, it assists in managing high-dimensional data by addressing linear relationships when representing text as vectors in NLP. Additionally, Quantum Amplitude Estimation (QAE) and Quantum Fourier Transform (QFT) are valuable for analyzing text features. QAE captures essential information within the text, while QFT helps to understand the overall structure and relationships within the data. These quantum algorithms offer promising improvements in both the efficiency and accuracy of text processing.

III. Methodology

A. Dataset Description and Preprocessing

In the study of Chinese text processing, this paper utilizes some useful datasets, which provide the necessary materials for the experiments. Among them, the Chinese Massive Multitask Language Understanding (CMMU) [9] is a comprehensive Chinese evaluation benchmark specifically designed to assess language models' knowledge and reasoning abilities in the Chinese context. This benchmark covers 67 topics, ranging from fundamental subjects to advanced professional levels, including natural sciences that require calculation and reasoning, humanities and social sciences that depend on knowledge, as well as everyday knowledge like Chinese driving rules. GaoKao-Bench [10] is a dataset based on China's Gaokao exam questions, aimed at evaluating large models' language comprehension and logical reasoning abilities. AGIEval [11] is designed to evaluate the general capabilities of foundational models in tasks related to human cognition and problem-solving.

B. Proposed Approach

This paper first provides a systematic review of the foundational concepts related to ML, QML, and long text processing, with a particular focus on understanding long Chinese texts. It introduces key algorithms in these fields and emphasizes the development and application of ML and QML techniques in this context. The core contribution of the study lies in comparing the performance of ML and QML in processing long Chinese texts, with a focus on processing efficiency and text similarity. Through experimental results, the paper demonstrates the advantages of QML in accelerating text processing, reducing time complexity, and efficiently extracting text features. Furthermore, the study explores the potential of QML in enhancing the ability of LLMs to process long Chinese texts, proposing a potential QML architecture and examining its integration into LLMs. Finally, the paper discusses the limitations of current research in this area and offers suggestions for future work. The overall structure of the paper is depicted in Fig. 2.

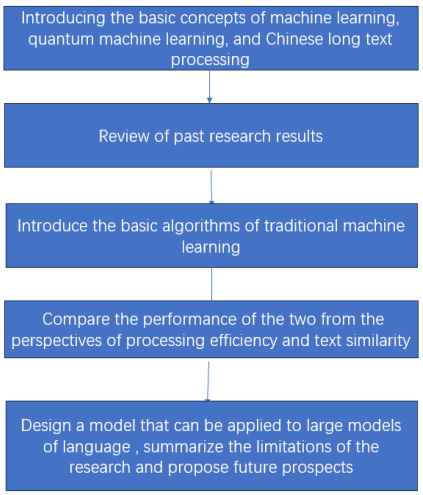


Figure 2. Overall Structure of the Paper.

1) ML

ML is the process of enabling computer systems to automatically learn from data and make predictions or decisions through algorithms and statistical models. According to actual needs, ML can be divided into supervised learning, unsupervised learning and reinforcement learning. According to actual needs,

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ML includes several typical algorithms, this paper only lists some examples.

Simple linear regression is exactly what it sounds like: it is a very straightforward simple linear regression method for predicting a quantitative response Y based on a single predictor variable X. It assumes that there is an approximate linear relationship between X and Y. Mathematically, this linear regression can be written as:

$$Y = \beta_0 + \beta_1 X \quad (1)$$

$e_i = y_i - \hat{y}_i$ is the i-th residual - this is the difference between the actual response value y_i and the predicted value $\hat{y}_i$ from the linear model. The residual sum of squares (RSS) is defined as:

$$\text{RSS} = \sum_{i=1}^{n} {e_i}^2 \quad (2)$$

Then, the gradient descent algorithm can be used to find the two coefficients that minimize the RSS, which are β_0 and β_1 to be solved [12].

For Support Vector Machine (SVM), in a feature space, a hyperplane is a plane that is used to separate data. In a binary classification problem, SVM looks for a hyperplane that separates data points of different categories. Support vectors are those data points that are closest to the hyperplane. These points are crucial to determining the hyperplane because if other points are removed, the definition of the hyperplane does not change. By focusing on these support vectors, SVM is able to build a classifier with good generalization ability. For linearly separable data, SVM can directly find a hyperplane to separate the data. However, in many practical applications, the data may be nonlinearly separable. To solve this problem, SVM introduces the concept of kernel function, which allows the data to be mapped to a higher-dimensional space. In a high-dimensional space, the originally nonlinearly separable data may become linearly separable.

2) QML

QML is an interdisciplinary field that combines quantum computing and ML In the quantum field, a quantum bit is described as a two-dimensional complex vector space, which contains two standard orthogonal bases, which represent the two possible states of the quantum bit:

$$|0>=[0,1]^T \quad |1>=[1,0]^T \quad (3)$$

Any single quantum bit state can be expressed as:$|\phi>=\alpha|0>+\beta|1>$, The coefficients α and βare complex numbers, called amplitudes, satisfying: $|\alpha|^2 + |\beta|^2 = 1$, This quantum state represented as a linear combination is also called a superposition state [13]. Quantum logic gates are the basic units of quantum computing, responsible for manipulating the state of quantum bits (qubits). Unlike classical logic gates, quantum logic gates can generate superposition and entangled states.The design and implementation of quantum logic gates is a key factor in the performance of quantum computers and further implementation of QML.

QML uses the high parallelism of quantum computing to further optimize traditional ML. From the perspective of information theory, it can be proved that the learnability of quantum information and classical information is equivalent, which further promotes the development of QML. QML algorithms generally include three steps: first, converting scenic spot information into quantum information; second, implementing the quantum version of the traditional ML algorithm; and finally, extracting the final calculation results. Since the calculation results are in quantum states and cannot be used directly, they need to be subjected to quantum measurement operations to collapse the quantum superposition state wave packet into a classical state and extract the classical information [14].

3) QML in Long Chinese Text Processing

The process of applying QML to long Chinese text processing begins with Quantum Encoding, where the text is transformed into a format that quantum computing can understand. This step is analogous to text encoding in classical ML, such as using word or sentence vectors. In QML, each word, character, or word vector is encoded as quantum state vectors and mapped to quantum bits (qubits). Common encoding methods include amplitude encoding and phase encoding, which allow different semantic units within the text to be stored in parallel in quantum states. Following this is Quantum Feature Extraction, where quantum computing enhances the efficiency of processing the complex dependencies found in long Chinese texts by accelerating the feature extraction process. Three primary algorithms are utilized during this step. QAE quickly estimates important features in the text, such as word frequency, key topics, or significant sentences. The QFT aids in understanding the global structure of the text by extracting useful frequency-related features. Additionally, the HHL algorithm solves linear equations via quantum computing, helping to identify eigenvectors and eigenvalues in matrices, which is useful for uncovering hidden patterns or topic distributions within the text.

Given that long Chinese texts often exhibit complex contextual dependencies, QML introduces a Quantum Attention Mechanism. By leveraging quantum superposition, the model can process multiple context windows simultaneously, enhancing its ability to focus on various text fragments efficiently. In classical NLP, self-attention mechanisms are used to calculate the correlation between words. In quantum self-attention, quantum entanglement is employed to process correlations between multiple positions in parallel, improving computational efficiency and reducing processing time. Finally, the quantum-enhanced language model is trained using a Quantum Optimization Algorithm. The acceleration capabilities of quantum computing significantly reduce training time, especially when dealing with large-scale text datasets. After training, the model can be applied for inference and generation of long Chinese texts, demonstrating QML's potential to improve both processing efficiency and semantic understanding.

IV. Result and Discussion

A. Performance Comparison Between ML and QML

To enhance the performance of large models in processing long Chinese texts using QML, two key approaches can be adopted: first, reducing the time required for large models to process the texts; and second, ensuring that the semantics of the processed text remain as consistent as possible with the original text.

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When traditional ML processes long Chinese texts, it first calculates the relevance of each word to all other words through the self-attention mechanism of models such as Transformer. Secondly, it optimizes weights and parameters through multiple iterations, and each iteration requires calculations on the entire text. This makes the time complexity of traditional ML when processing long Chinese texts reach $O(n^2)$ (n is the length of the text) or even higher. The QML model makes full use of the advantages of quantum computing. First, it can use quantum parallelism to perform parallel calculations between multiple states to avoid calculating the features of each character of the text one by one. Secondly, because quantum states allow multiple inputs to be processed simultaneously, it is no longer necessary to calculate each character independently when processing long Chinese texts. Ultimately, the complexity of QML can be reduced to $O(\log n)$ or $O(\sqrt{n})$Design a python algorithm to simulate the time complexity of traditional ML and QML respectively, run it 2 times, and get the two figures : Fig. 3, Fig. 4.

In Fig. 3 and Fig. 4, the horizontal axis represents the number of Chinese long text characters processed, and the vertical axis represents the processing time. The blue line is the processing time of traditional ML, and the orange line is the processing time of QML. It can be seen that the time for QML to process text is significantly lower than that of traditional ML. The superposition and entanglement of quantum computing make it possible to represent data in a higher-dimensional space. Quantum states allow multiple information to be stored on qubits at the same time, potentially able to capture complex associations in text data. For the language structures with strong semantic and contextual dependencies in long Chinese texts, QML can provide richer feature representations, thereby better understanding the text content.

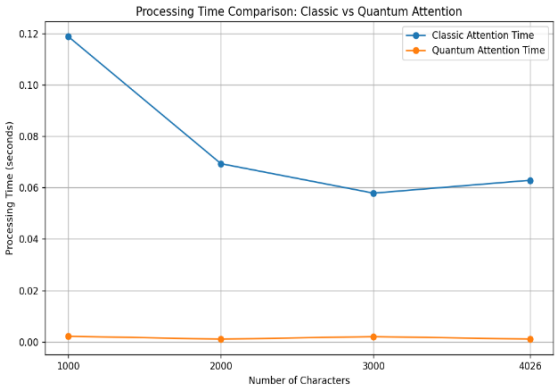


Figure 3. The First Comparison between Classic and Quantum Attention Time

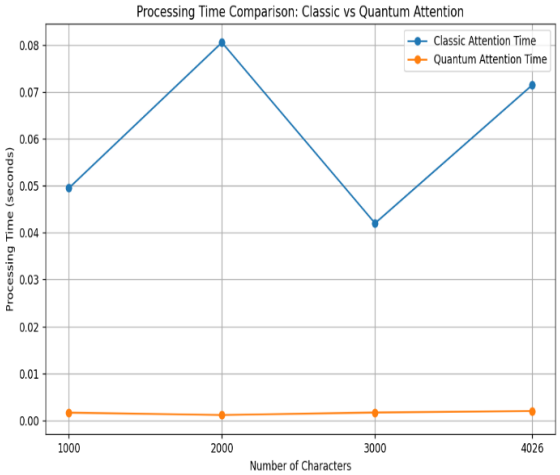


Figure 4. The Second Comparison between Classic and Quantum Attention Time

B. *Model Design and Discussion*

To effectively apply QML technology for processing long Chinese texts within LLMs, a simple model consisting of five layers was designed. The first layer, the Quantum Embedding Layer, distinguishes itself from traditional embedding layers by not only mapping words into high-dimensional space but also incorporating quantum superposition, allowing each word to be represented as a superposition of multiple states. Next, the Quantum Self-Attention Mechanism Layer leverages the parallel computing capabilities of quantum states to rapidly compute the similarities and relevance among different words. The Quantum Entanglement Layer captures long-distance dependencies between Chinese words by utilizing the properties of quantum entanglement. Following this, the Decoherence and Feature Compression Layer refines the high-dimensional information into low-dimensional features through decoherence, streamlining the data for further processing. Finally, the Output Layer employs the classic Transformer output mechanisms, such as classification, generation, and translation, to produce the final results. This layered architecture improves the model's ability to efficiently process and understand long Chinese texts, showcasing the potential of QML in this task.

When faced with the task of processing long Chinese texts, QML can use its unique properties to speed up calculations, enhance text feature representation capabilities, and better understand texts compared to traditional ML. When applied to LLM, it can better process texts sent by users and make answers that better meet user requirements. However, the model designed in this paper to apply the technology of QML to process long Chinese texts to LLM is only a theoretical design. In actual operation, it is limited by quantum hardware conditions. In addition, the effective integration of quantum feature extraction and classical language models requires new algorithm design and system architecture, and may face challenges in compatibility and performance optimization. These all need to be studied in the future.

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V. CONCLUSION

In light of the challenges faced by LLMs in processing long Chinese texts—such as the high complexity of the language, its rich semantics, and the inherent difficulty in achieving accurate understanding—this paper proposes a novel approach to apply QML to address these issues. The paper begins by reviewing the recent advancements in QML, followed by an explanation of the basic concepts and algorithms underlying both traditional ML and QML. It then details the principles of applying QML to the processing of long Chinese texts. The study conducts a comparative analysis of ML and QML from two key perspectives: text processing time and comprehension accuracy. Through this analysis, the advantages and limitations of each approach are highlighted. The results demonstrate that QML outperforms traditional ML in terms of both processing efficiency and the degree of understanding achieved when handling long Chinese texts. In addition, a simplified QML-based model for integration with LLMs is proposed, offering a promising avenue for improving LLM performance. Looking ahead, the practical implementation and constraints of applying QML to LLMs will need to be carefully considered, and further research is important to explore the full potential of QML in this domain.

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